SQL ASSIGNMENT – 3

GenAI Application – SQL Data Types and Constraints

1. Create a database named GenAI_App
2. Create a table named Users with the following columns:
   - user_id – Integer, Primary Key
   - username – Maximum 50 characters, cannot be NULL
   - email – Maximum 100 characters, must be unique
   - age – Integer, must be 18 or above
   - phone – Fixed 10 characters
   - registration_date – Date
   - account_status – Default value should be 'Active'
3. Create a table named AI_Models with:
   - model_id – Integer, Primary Key
   - model_name – Maximum 50 characters, cannot be NULL
   - model_version – Maximum 20 characters
   - is_available – Boolean, default value TRUE
4. Create a table named Conversations with:
   - chat_id – Integer, Primary Key
   - user_id – Integer, Foreign Key referencing Users
   - model_id – Integer, Foreign Key referencing AI_Models
   - user_prompt – Text, cannot be NULL
   - ai_response – Text
   - token_count – Integer, must be zero or above
   - chat_time – Date and Time
5. create a table named payments with:
   - Payment_id – int,primary key
   - User_id – int, foreign key referencing Users
   - Amount- decimal
   - Payment date – date and time
   - Payment_status
6. Display all records from the four tables.
7. Display only the username and email of all users.
8. Display users whose age is greater than 25.

9. Display AI models that are currently available.

10. Display conversations where token_count is greater than 500.

11. Display payments with an amount greater than ₹500.

12. Try inserting a duplicate email into the Users table. Observe the result.

13. Try inserting a user below 18 years of age. Observe the result.

14. Insert a payment without specifying payment_status. Check the default value.

Answers:

1. Create a database named GenAI_App

   Query:

```
create database GenAI_App;
use GenAI_App;
```

2.
```
-- 2. Create a table named Users with the following columns:
-- user_id – Integer, Primary Key
-- username – Maximum 50 characters, cannot be NULL
-- email – Maximum 100 characters, must be unique
-- age – Integer, must be 18 or above
-- phone – Fixed 10 characters
-- registration_date – Date
-- account_status – Default value should be 'Active
```

   Query:

```
create table Users(
User_ID int primary key,
Username varchar(50) Not null,
Email varchar(50) unique,
```

```
Age int check(Age > 18),
Phone varchar(10),
Registration_date date,
Account_status varchar(10) default 'Active');

-- Data insertion --
insert into Users values
(1001,"Prabanchan","Prabaarul@email.com",21,"9952470784","2026-10-
21",default),
(1002,"krishna","krishna@email.com",23,"9056789234","2025-09-
25","inactive"),
(1003,"Arul","arul@email.com",52,"7339413422","2024-12-12",default),
(1004,"Aishwarya","aishu@gmail.com",21,"8956784509","2026-07-
20","suspended"),
(1005,"Latha","latha@gmail.com",45,"9025678921","2023-05-
21",default);
```

3. Create a table named AI_Models with:
   -- model_id – Integer, Primary Key
   -- model_name – Maximum 50 characters, cannot be NULL
   -- model_version – Maximum 20 characters
   -- is_available – Boolean, default value TRUE

   Query:

```
create table AI_Models(
Model_ID int primary key,
Model_Name varchar(50) not null,
Model_version varchar(20),
Is_Available boolean default TRUE);

-- Data Insertion
insert into AI_Models values
(1111,"Gemini-Flash","v3.0",Default),
```

(1112,"Claude-Sonnet","v6.0",False),
(1113,"Mixtral-8x22B","v5.0",default),
(1114,"GPT-4o","v3.0",False);

4. Create a table named Conversations with:
   -- chat_id – Integer, Primary Key
   -- user_id – Integer, Foreign Key referencing Users
   -- model_id – Integer, Foreign Key referencing AI_Models
   -- user_prompt – Text, cannot be NULL
   -- ai_response – Text
   -- token_count – Integer, must be zero or above
   -- chat_time – Date and Time

   Query:
   create table Conversations(
   Chat_ID int primary key,
   user_ID int not null,
   model_ID int not null,
   User_prompt text not null,
   AI_response text,
   token_count int check(token_count >= 0),
   chat_time datetime,

   foreign key(user_ID)
   references Users(User_ID), -- User table User ID reference

   foreign key(model_ID)
   references AI_models(Model_ID)); -- AI_Model Model_ID reference

   -- Data Insertions --
   insert into Conversations values
   (1011,1001,1113,"Fix this grammar: She don't know nothing.","She doesn't know anything.",250,"2026-07-23 12:05:00"),

(1012,1004,1111,"Generate a quick JavaScript arrow function to square a number.","const square = x => x * x;",640,"2026-07-23 18:09:00"),
(1013,1003,1112,"What does the SQL command TRUNCATE do?","It deletes all rows from a table quickly without logging individual row deletions.",310,"2026-07-23 20:07:00"),
(1014,1002,1113,"Give a quick tip to reduce stress.","Try the 4-7-8 breathing technique: inhale for 4 seconds, hold for 7, and exhale for 8.",780,"2026-07-23 09:08:00");

5. create a table named payments with:
    - Payment_id – int,primary key
    - User_id – int, foreign key referencing Users
    - Amount- decimal
    - Payment date – date and time
    - Payment_status

    Query:

    CREATE TABLE Payments (

    payment_id INT PRIMARY KEY,

    user_id INT,

    amount DECIMAL(10,2),

    payment_date DATETIME,

    payment_status VARCHAR(20) default 'Paid',

    FOREIGN KEY (user_id) REFERENCES Users(user_id));

    -- Data Insertion --

    insert into Payments values

    (1101,1001,250.00,"2026-03-23 12:45:00",default),

    (1102,1002,650.00,"2026-10-10 15:00:00","unpaid"),

(1103,1004,1000.00,"026-11-12 23:00:45",default),

(1104,1005,600.00,"2026-10-07 14:00:00","processing");

6. Display all records from the four tables.

Query:

Select * from Users;

Output:

User_ID	Username	Email	Age	Phone	Registration_date	Account_status
1001	Prabanchan	Prabaarul@email.com	21	9952470784	2026-10-21	Active
1002	krishna	krishna@email.com	23	9056789234	2025-09-25	inactive
1003	Arul	arul@email.com	52	7339413422	2024-12-12	Active
1004	Aishwarya	aishu@gmail.com	21	8956784509	2026-07-20	suspended
1005	Latha	latha@gmail.com	45	9025678921	2023-05-21	Active
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Query:

select * from AI_Models;

Output:

Model_ID	Model_Name	Model_version	Is_Available
1111	Gemini-Flash	v3.0	1
1112	Claude-Sonnet	v6.0	0
1113	Mixtral-8x22B	v5.0	1
1114	GPT-4o	v3.0	1
NULL	NULL	NULL	NULL

Query:

select * from Conversations;

Output:

Chat_ID	user_ID	model_ID	User_prompt	AI_response	token_count
1011	1001	1113	Fix this grammar: She don't know nothing.	She doesn't know anything.	250
1012	1004	1111	Generate a quick JavaScript arrow function to s...	const square = x => x * x;	640
1013	1003	1112	What does the SQL command TRUNCATE do?	It deletes all rows from a table quickly without l...	310
1014	1002	1113	Give a quick tip to reduce stress.	Try the 4-7-8 breathing technique: inhale for 4 ...	780
NULL	NULL	NULL	NULL	NULL	NULL

Query:

select * from Payments;

Output:

payment_id	user_id	amount	payment_date	payment_status
1101	1001	250.00	2026-03-23 12:45:00	Paid
1102	1002	650.00	2026-10-10 15:00:00	unpaid
1103	1004	1000.00	0026-11-12 23:00:45	Paid
1104	1005	600.00	2026-10-07 14:00:00	processing
NULL	NULL	NULL	NULL	NULL

7. display only the username and email of all users.

Query:

select Username,Email

from Users;

Output:

Username	Email
Prabanchan	Prabaarul@email.com
krishna	krishna@email.com
Arul	arul@email.com
Aishwarya	aishu@gmail.com
Latha	latha@gmail.com

8. Display users whose age is greater than 25.

Query:

select * from users

where Age > 25;

Output:

User_ID	Username	Email	Age	Phone	Registration_date	Account_status
1003	Arul	arul@email.com	52	7339413422	2024-12-12	Active
1005	Latha	latha@gmail.com	45	9025678921	2023-05-21	Active
NULL	NULL	NULL	NULL	NULL	NULL	NULL

9. Display AI models that are currently available.

   Query:
   select * from AI_Models
   where Is_Available = TRUE;

   Output:

Model_ID	Model_Name	Model_version	Is_Available
1111	Gemini-Flash	v3.0	1
1113	Mixtral-8x22B	v5.0	1
NULL	NULL	NULL	NULL

10. Display conversations where token_count is greater than 500.

    Query:
    select chat_ID,User_ID,Model_ID,Token_count from Conversations
    where Token_count > 500;

    Output:

chat_ID	User_ID	Model_ID	Token_count
1012	1004	1111	640
1014	1002	1113	780

11. Display payments with an amount greater than ₹500.

    Query:
    select * from payments
    where amount > 500.00;

    Output:

payment_id	user_id	amount	payment_date	payment_status
1102	1002	650.00	2026-10-10 15:00:00	unpaid
1103	1004	1000.00	0026-11-12 23:00:45	Paid
1104	1005	600.00	2026-10-07 14:00:00	processing
NULL	NULL	NULL	NULL	NULL

12. Try inserting a duplicate email into the Users table. Observe the result.

    Query :
    insert into Users values
    (1006,"Saran","prabaarul@email.com",21,"9345193361","2026-11-10",default);

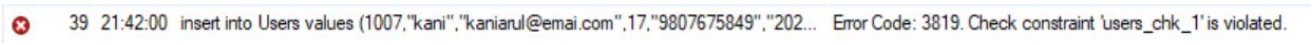
    Output:



13. Try inserting a user below 18 years of age. Observe the result.

    Query:
    insert into Users values
    (1007,"kani","kaniarul@emai.com",17,"9807675849","2026-11-11","inactive");

    Output:



14. Insert a payment without specifying payment_status. Check the default value

    Query:
    insert into payments (payment_id,user_id,amount,payment_date) values
    (1105,1003,1500.00,"2026-11-10 12:00:00");
    select * from payments;
    Output:

payment_id	user_id	amount	payment_date	payment_status
1101	1001	250.00	2026-03-23 12:45:00	Paid
1102	1002	650.00	2026-10-10 15:00:00	unpaid
1103	1004	1000.00	0026-11-12 23:00:45	Paid
1104	1005	600.00	2026-10-07 14:00:00	processing
1105	1003	1500.00	2026-11-10 12:00:00	Paid
NULL	NULL	NULL	NULL	NULL